SWITCHING THE ORDER IN THE DETERMINANT DET(ID + A B) = DET(ID + B A) FOR MAPS BETWEEN FINITE PROJECTIVE MODULES OVER ANY RING

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Allowed standard prerequisites (editorial).

- An equality of elements of a ring of the form det(id + a b) = det(id + b a) may be checked after localizing at every prime, and over a local ring a finite projective module is finite free. used at proof lines 36431-36433, where the author replaces R by a localization at a prime ideal and then assumes R local with M and N finite free. Reduces the statement for arbitrary finite projective modules over an arbitrary ring to the matrix identity of the next step, over a commutative ring of the simplest kind.
- In characteristic zero the determinant of a square matrix of size at most N is a universal polynomial in the traces of the powers of that matrix. invoked at proof lines 36442-36445, where the author passes to the fraction field of the polynomial ring and then uses that a universal polynomial expresses the determinant in the traces of the powers. Converts the matrix identity for the two rectangular matrices AB and BA into the family of trace identities Trace((I_n + AB)^k) = Trace((I_m + BA)^k), which is what can be expanded.

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- The trace identity Trace(XY) = Trace(YX) for matrices of compatible size, and the factorizations (AB)^k = A(BA)^{k-1}B and (BA)^k = (BA)^{k-1}AB. used at proof lines 36451-36454. Supplies the last step: the traces of all powers agree, hence by the previous item the two determinants agree.

Lemma 0.2. *Let R be a ring. Let M, N be finite projective R-modules. Let $a : M \to N$ and $b : N \to M$ be R-linear maps. Then*

$$\det(id + a \circ b) = \det(id + b \circ a)$$

as elements of R.

Proof. It suffices to prove the assertion after replacing R by a localization at a prime ideal. Thus we may assume R is local and M and N are finite free. In this case we have to prove the equality

$$\det(I_n + AB) = \det(I_m + BA)$$

of usual determinants of matrices where A has size $n \times m$ and B has size $m \times n$. This reduces to the case of the ring $R = \mathbf{Z}[a_{ij}, b_{ji}; 1 \leq i \leq n, 1 \leq j \leq m]$ where a_{ij} and b_{ij} are variables and the entries of the matrices A and B. Taking the fraction field, this reduces to the case of a field of characteristic zero. In characteristic zero there is a universal polynomial expressing the determinant of a matrix of size $\leq N$ in the traces of the powers of said matrix. Hence it suffices to prove

$$\text{Trace}((I_n + AB)^k) = \text{Trace}((I_m + BA)^k)$$

for all $k \geq 1$. Expanding we see that it suffices to prove $\text{Trace}((AB)^k) = \text{Trace}((BA)^k)$ for all $k \geq 0$. For $k = 1$ this is the well known fact that $\text{Trace}(AB) = \text{Trace}(BA)$. For $k > 1$ it follows from this by writing $(AB)^k = A(BA)^{k-1}B$ and $(BA)^k = (BA)^{k-1}AB$. □